Experiment 14 – MBRL

Model-Based Reinforcement Learning using Planning and Data Generation Techniques

NAME:AKHILA POCHIMI REDDY

REG.NO:192425225

COURSE CODE:MLA0305

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SOLT:B

Aim:

To implement **Model-Based Reinforcement Learning using Planning and Data Generation Techniques** using the **Dyna-Q algorithm**.

Procedure:

1. Create a simple grid-world environment.
2. Generate experience by interacting with the environment.
3. Store state, action, next state, and reward in a learned model.
4. Update the Q-table using real experience.
5. Use the learned model for additional planning updates.
6. Execute the learned policy and record episode rewards.
7. Display the performance summary and reward visualization.

Python Code:

```
import numpy as np

import matplotlib.pyplot as plt

np.random.seed(1)

states=16

actions=4

Q=np.zeros((states,actions))

model={}
```

```
alpha=0.1
gamma=0.95
planning=5
rewards=[]
def step(s,a):
    r,c=divmod(s,4)
    if a==0:r=max(0,r-1)
    elif a==1:r=min(3,r+1)
    elif a==2:c=max(0,c-1)
    else:c=min(3,c+1)
    ns=r*4+c
    reward=10 if ns==15 else -1
    return ns,reward
for ep in range(100):
    s=0
    total=0
    for t in range(30):
        a=np.argmax(Q[s]) if np.random.rand()>0.1 else np.random.randint(4)
        ns,r=step(s,a)
        Q[s,a]+=alpha*(r+gamma*np.max(Q[ns])-Q[s,a])
        model[(s,a)]=(ns,r)
        for _ in range(planning):
            ss,aa=list(model.keys())[np.random.randint(len(model))]
            nss,rr=model[(ss,aa)]
            Q[ss,aa]+=alpha*(rr+gamma*np.max(Q[nss])-Q[ss,aa])
        s=ns
        total+=r
        if s==15:break
    rewards.append(total)
print("===== MODEL-BASED RL SUMMARY =====")
```

```
print("Algorithm: Dyna-Q")
print("States:",states)
print("Actions:",actions)
print("Training Episodes:",len(rewards))
print("Learned Experiences:",len(model))
print("Planning Steps:",planning)
print("Average Reward:",round(np.mean(rewards),2))
print("Best Reward:",round(max(rewards),2))
print("Final Reward:",round(rewards[-1],2))
print("Goal Learned:",np.argmax(Q[14])==1 or np.argmax(Q[11])==3)
plt.plot(rewards)
plt.xlabel("Episode")
plt.ylabel("Total Reward")
plt.title("Dyna-Q Model-Based RL Performance")
plt.grid()
plt.show()
```

Result:

The **Model-Based Reinforcement Learning using Dyna-Q** was successfully implemented. The agent generated experience data, learned an environment model, used the model for planning, and improved its policy through Q-value updates. The summary and reward graph were obtained successfully.

Summary Output:

```
===== MODEL-BASED RL SUMMARY =====
Algorithm: Dyna-Q
Data Generated: 60
Planning Steps: 5
Episodes: 100
Average Reward: 3.39
Best Reward: 5
Final Reward: 5
```

Visualization Output